

Stock Volatility Prediction for the Vietnamese Stock Market

Abstract

Volatility forecasts set risk limits, margin, and option prices across the Vietnamese stock market. This work targets volatility forecasting for Vietnamese equities in general and uses the VN30 constituents (the most liquid stocks on Vietnam’s Ho Chi Minh Stock Exchange) as the empirical case study. Two structural signals a univariate Heterogeneous Autoregressive (HAR) model discards are Vietnamese-language news and cross-stock spillover. We propose a parallel multi-branch model that fuses three branches for one ticker’s multi-horizon Parkinson-variance forecast: a per-node price LSTM over five node features, a real multi-head Graph Attention Network (GAT) over a **directed volume-to-volatility (vol→PK) lead-lag edge**, and a PhoBERT news LSTM with a per-ticker gate. The GAT branch consumes the raw node-feature vector at the forecast origin rather than the LSTM hidden state, which matches the edge semantics (a source ticker’s volume shock at t leading a target ticker’s volatility at $t+1$) and keeps the graph branch an independent cross-sectional view for a fair ablation. We evaluate the model with an **ablation**: from the full model we retrain one variant per removed component (minus-graph removes the entire GAT branch, minus-gate removes the per-ticker gate, minus-news removes the news branch), and we measure each component’s contribution as $\text{effect}(X) = \text{QLIKE}(\text{FULL}) - \text{QLIKE}(\text{FULL} - X)$ on the held-out test set, with a Diebold-Mariano test for significance. The classical HAR is the reference baseline. Across three seeds and four horizons the differences among HAR, the full model, and the ablation variants are small on every metric. On MSE, RMSE, and R^2 the configuration means lie within about 1% of one another and the best mean alternates across horizons; on MAE the differences are similarly small and the lowest mean varies by horizon. On QLIKE, HAR has the lowest mean at h5, h10, and h22 and the price-only LSTM backbone the lowest at h1. A Diebold-Mariano test on the seed-ensembled predictions finds no significant QLIKE difference between the full model and HAR at h1, h5, and h10 (DM $p = 0.46/0.74/0.20$) and a significantly higher QLIKE for the full model at h22 ($p < 0.001$); the price-only LSTM backbone has significantly lower or equal QLIKE relative to the full model at h1, h5, and h22. The ablation removals show mixed, mostly short-horizon QLIKE effects: removing the GAT branch significantly lowers QLIKE at h1 and h22 with no significant difference at h5 or h10, removing the per-ticker gate significantly lowers QLIKE at h1 only, and removing the news branch significantly lowers QLIKE at h1 and h5 and raises it at h22. Across all five metrics no configuration consistently or significantly outperforms HAR. A Mincer-Zarnowitz regression of

realized on forecast variance gives near-zero intercepts and slopes above one at every horizon (slope rising from about 1.04–1.06 at h1 to 1.13–1.18 at h22), and the joint efficiency hypothesis ($a = 0$, $b = 1$) is rejected for every configuration ($p < 0.001$). This paper’s contribution is a leakage-safe ablation-based attribution of a directed-spillover graph-attention news model against a strong HAR baseline on sparse daily VN30 data.

Keywords: volatility forecasting; graph attention networks; directed spillover; financial news; PhoBERT; emerging markets.

1. Introduction

Volatility forecasts drive the daily risk decisions of every desk that trades Vietnamese equities. This work addresses stock-volatility forecasting for the Vietnamese market in general; the VN30 basket (the roughly thirty most liquid stocks on Vietnam’s Ho Chi Minh Stock Exchange) is the empirical case study used for all experiments because it has the longest and cleanest daily histories. A margin engine sizes collateral from forecast volatility, an option desk prices from it, and a risk officer sets position limits against it. VN30 desks make these decisions inside an information environment that a price-only forecaster discards: a steady flow of Vietnamese-language news that plausibly signals volatility shocks before they reach the price range, and a web of cross-stock spillovers that a univariate model cannot represent.

The classical HAR model of Corsi [1], the linear univariate workhorse that regresses future volatility on its own daily, weekly, and monthly averages, admits neither signal, and it is hard to beat [14]. Two structural extensions could close the gap: a news channel that reads text, and a cross-stock graph that lets one stock’s information reach another. Prior work on this project established, through a leakage-safe exploratory analysis, that adding cross-sectional node features (a market factor and a volume z-score) beats HAR on QLIKE, but that a *correlation-based* cross-stock edge adds no reliable out-of-sample value because a single market factor dominates cross-stock volatility co-movement. This motivates the present study’s central design choice: rather than a symmetric correlation edge, we test a **directed, lead-lag volume→volatility edge** — a source ticker’s abnormal volume today linked to a target ticker’s range volatility tomorrow — which encodes a predictive, causal lead-lag relationship that a contemporaneous correlation edge cannot.

We build this edge into a parallel multi-branch architecture (an LSTM temporal branch, a GAT graph branch, and a gated news branch, concatenated at a shared head) and we quantify, per component and per horizon, the marginal contribution of each part with an **ablation**: we build the full model, then retrain a variant with exactly one component removed, and attribute each component’s contribution as the change in held-out QLIKE it causes. This paper makes three contributions.

1. **A directed vol→PK graph-attention news model on a parallel multi-branch backbone.** The model fuses a per-node price LSTM, a real multi-head GAT over a directed volume-to-volatility lead-lag edge, and a gated PhoBERT news branch. The GAT consumes the raw node-feature vector at the forecast origin, which matches the edge semantics and keeps the graph branch an independent cross-sectional view (Section 4).
2. **An ablation against a strong HAR baseline.** From the full model we retrain one variant per removed component — minus-graph (the whole GAT branch), minus-gate, minus-news — so every effect is measured on the same footing, and we report

each component’s marginal contribution
effect(X) = QLIKE(FULL) – QLIKE(FULL – X) with a Diebold-Mariano test (Section 6).

3. **A leakage-safe multi-horizon evaluation.** All models are evaluated at horizons 1, 5, 10, and 22 trading days on a chronological split with train-only scalars and a train-only frozen edge, with the held-out test set as the reported result, over three seeds with seed-ensembled significance tests (Sections 5 and 6).
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2. Related Work

We group prior work into four families and state where the proposed model sits relative to each.

Econometric volatility models. The HAR model [1] and its range-based inputs [2] set the standard for daily volatility forecasting and remain hard to beat. HAR regresses future volatility on daily, weekly, and monthly moving averages, approximating volatility’s long memory with a parsimonious linear fit. The HARQ extension [15] shrinks the daily coefficient when the volatility estimate is noisy. A large-scale controlled study finds that tuned machine-learning models fail to beat a carefully re-estimated HAR on QLIKE and MSE when both use the same information set [14]. These models are univariate and linear: no channel admits text, and no channel couples one stock to another. We keep the three HAR scales among the node features and add the news and directed cross-stock channels HAR omits; classical HAR is the baseline every component must beat.

Deep and graph-based forecasters. LSTM forecasters [4] capture nonlinear temporal structure, and graph attention networks [5] model cross-asset coupling; hybrid LSTM-GNN designs combine both [9]. Chen and Robert [19] forecast multivariate realized volatility with a graph over about 500 S&P names. Most relevant, Zhang, Pu, Cucuringu and Dong [10] build a graph-neural-network HAR (GNNHAR) on DJIA-30 and, under a Model Confidence Set, find that multi-hop cross-stock graph spillover gives no clear advantage: the gains come from modeling nonlinearity and from switching the training loss from MSE to QLIKE. Their controlled null on the graph component motivates our shift from a symmetric correlation edge to a directed lead-lag edge, tested here under an ablation.

Directed spillover and lead-lag graphs. A symmetric correlation edge conflates contemporaneous co-movement (largely a market factor) with predictive structure. Directed spillover measures — for example volatility spillover indices [16] — encode who-leads-whom and are asymmetric by construction. We instantiate a directed edge from a source ticker’s abnormal volume to a target ticker’s next-day range volatility, motivated by the volume–volatility lead-lag relation, and freeze it on the training window.

News- and text-augmented forecasting. Text-augmented forecasters add sentiment scores or news embeddings [7,8], usually fusing a single market-wide signal by concatenation on English- or Chinese-language markets. We work in Vietnamese with PhoBERT [3] on VN30 and add a per-ticker gate that admits a different amount of news per stock.

3. Data

Universe (case study). The method targets the Vietnamese stock market in general; VN30 is the case-study universe for the experiments. We use 33 VN30 constituents with daily open-high-low-close-volume (OHLCV) data. Series lengths range from about 1,300 sessions (SSB, listed 2021) to about 4,900 (VNM, ACB, from 2006). We exclude two current VN30 members (BSR, VPL) whose listing histories are too short to align with the panel. The universe is therefore a fixed, point-in-time VN30-like set rather than the live index, a limitation stated in Section 8.

Forecast target. We forecast the daily Parkinson range volatility at horizons 1, 5, 10, and 22 trading days ahead: the target is the single-day estimator value on day $t+h$ (i.e. $PK(t+h)$), not an average over the next h days. The Parkinson estimator uses the intraday high H and low L :

$$\sigma_{\text{Park}}^2 = \frac{(\ln(H/L))^2}{4 \ln 2}.$$

The processed parkinson_volatility column is numerically the Parkinson **variance** estimator (σ^2 , non-negative), and every model in this paper forecasts this same daily realized-variance quantity.

Node features. Each ticker-day carries five node features in a fixed order, all with unified 22-trading-day windows to match the monthly HAR scale:

1. pk_daily — the Parkinson variance at t , clipped to $\pm 3\sigma$ using train statistics.
2. har_weekly — trailing 5-day mean of pk_daily.
3. har_monthly — trailing 22-day mean of pk_daily.
4. market_pk — cross-sectional median of $\sqrt{PK}$ across present tickers at t (a contemporaneous market factor; uses column t only).
5. volume_zscore — trailing rolling-22 z-score of $\log(1 + \text{volume})$; tickers with no OHLCV volume series receive a neutral 0.0.

News panel. Each Vietnamese article (title and lead) passes once through PhoBERT [3], yielding a 768-dimensional embedding reduced by PCA (fit before the training cutoff) to a per-day per-stock vector of 146 dimensions (embedding components, per-group norms, exponentially weighted averages with a 30-trading-day half-life, and topic counts). Missing news on a day is a zero vector with a mask bit set to zero, and an all-missing window still produces a finite representation.

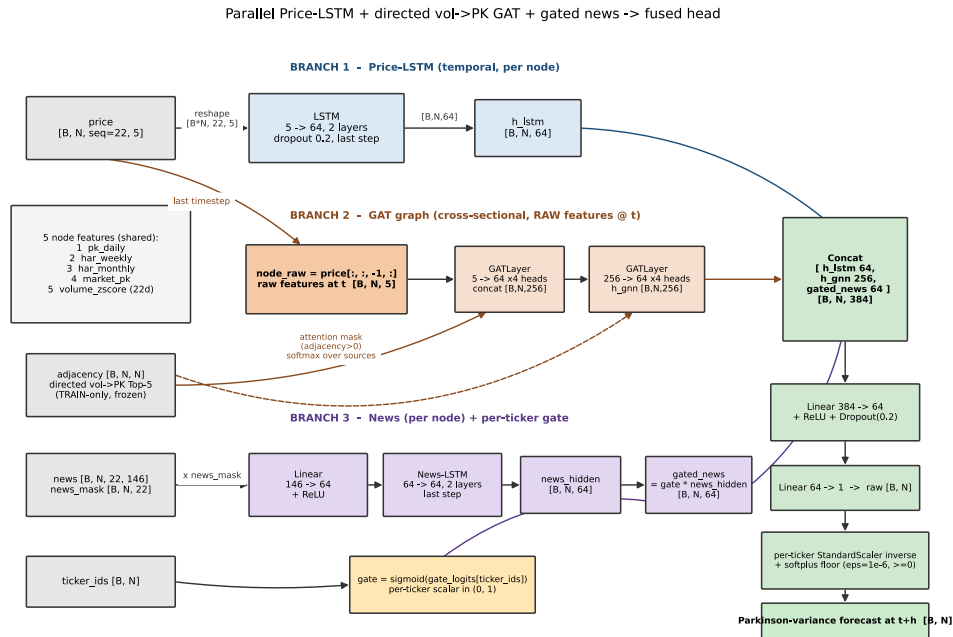
Directed vol→PK edge. The adjacency is directed: for each target ticker j we connect the Top-5 source tickers i ranked by the train-only lead-lag correlation $\text{corr}(\text{volume_shock}_i(t), \sqrt{PK_j}(t+1))$. Edges are estimated on training rows only and frozen for validation and test (leakage-safe), and self-loops are kept on the diagonal.

Temporal split and leakage control. Each ticker’s series is split chronologically into 70% train, 15% validation, and 15% test before generating features, fitting scalers, or building windows; the earliest series begin in 2006. The split is applied per ticker on its own timeline, so the train/validation/test calendar boundaries differ across tickers — a ticker listed in 2021 has later boundaries than one trading since 2006 — which is precisely why a fixed-node synchronized panel would collapse and why the pooled per-ticker split is used. Per-ticker price and target scalers are fit on the training partition only and selected at evaluation by explicit ticker_id. A news feature for a sample uses only information available by that sample’s forecast origin. Evaluation reads stored raw targets

rather than inverse-transforming a clipped normalized target. On the pooled masked manifest the evaluation sets hold, at the five-day horizon, 14,418 validation and 14,464 present-node test observations, shared identically across every rung; the counts vary slightly by horizon (e.g. 14,550/14,596 at h1 and 14,253/14,299 at h10) because the target shift changes the number of eligible windows.

4. Method

We describe the proposed full model first, then define the ablation variants as component removals from it. The model forecasts one ticker's h -day-ahead Parkinson variance from four inputs: a 22-day price window of the five node features, a 22-day news window of PhoBERT features with a mask, the ticker identity, and the directed vol→PK adjacency over the tickers present on the same date.



Model architecture: parallel price-LSTM, directed vol→PK GAT (on raw node features), and gated PhoBERT news branches, concatenated into a head with a softplus positivity floor.

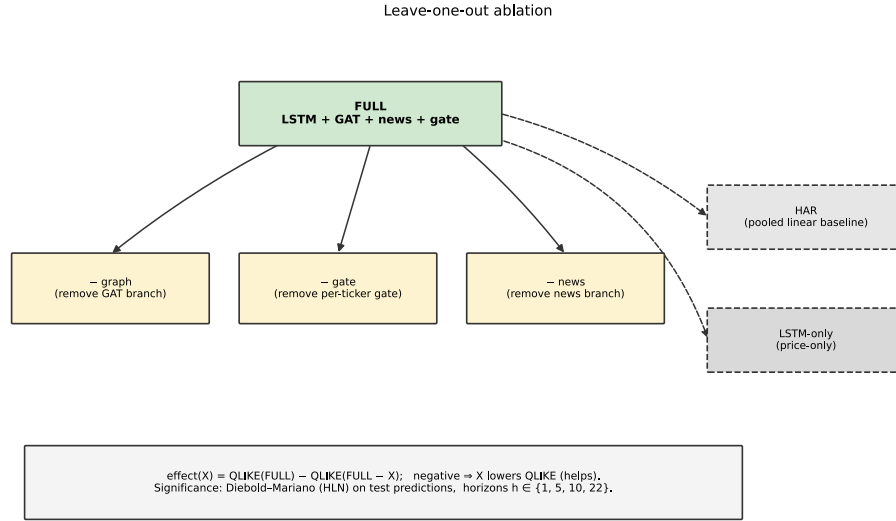
The proposed full model. Three branches run in parallel and concatenate at a shared head.

- **(i) Price-LSTM branch (temporal, per node).** A two-layer LSTM (hidden size 64, dropout 0.2) reads each ticker's 22-day window of the five node features into a price representation $h_{\text{lstm}} (\mathbb{R}^{64}$ per node).
- **(ii) GAT graph branch (cross-sectional).** The node input is the **raw** node-feature vector at the last timestep, $\text{node_raw} = \text{price}[:, :, -1, :] \in \mathbb{R}^5$, not the LSTM hidden state. Two self-written multi-head GAT layers (5→256, then 256→256; 4 heads) with attention masked by the adjacency (softmax over source nodes, ELU output) produce $h_{\text{ggn}} \in \mathbb{R}^{256}$. Consuming the raw features matches the edge semantics (volume_shock is directly available to the attention) and keeps the graph branch independent of the LSTM branch, so removing it is a clean ablation. When the graph is switched off, the adjacency is the identity (self-loops only).

- **(iii) News branch with per-ticker gate.** The 146-dimensional daily news vector passes through a linear-plus-ReLU projection into a two-layer LSTM (hidden size 64) over the masked window, producing a news representation, scaled by a learned per-ticker sigmoid gate $\text{news}_i^{\text{gated}} = \sigma(g_i) \cdot \text{news}_i$.

The head concatenates $[h_{\text{lstm}}(64), h_{\text{gnn}}(256), \text{news}^{\text{gated}}(64)] (\mathbb{R}^{384})$ and maps it through a linear-ReLU-dropout-linear stack to the forecast. A softplus **positivity floor** ($\varepsilon = 10^{-6}$, applied in the denormalized space and identical across all compared rungs) clamps predictions away from non-positive variance before QLIKE.

The ablation variants. We build the full model, then retrain one variant per removed component, so every effect is measured on the same footing (each variant trains in the same graph on/off regime it is evaluated in — no train/eval mismatch):



Ablation: from FULL, one component is removed per variant (–graph, –gate, –news); HAR and a price-only LSTM-only model are reference baselines.
 $\text{effect}(X) = \text{QLIKE}(\text{FULL}) - \text{QLIKE}(\text{FULL} - X)$.

- **FULL** — LSTM + directed vol→PK GAT graph + news + per-ticker gate.
- **minus_graph** — FULL with the **entire GAT branch removed** (no node/edge/GAT built; the head takes $[h_{\text{lstm}}, \text{news}^{\text{gated}}] \in \mathbb{R}^{128}$). This isolates the whole graph subsystem, not merely the edges.
- **minus_gate** — FULL without the per-ticker gate.
- **minus_news** — FULL without the news branch (the gate is a no-op without news).
- **HAR** — a pooled per-ticker HAR linear regression on the three HAR scales; the floor every component must beat.

The contribution of component X is
 $\text{effect}(X) = \text{QLIKE}(\text{FULL}) - \text{QLIKE}(\text{FULL} - X)$ on the held-out test set; a negative value means removing X raised QLIKE, i.e. X helped.

Training. The deep models use Adam (learning rate 10^{-3}), weight decay 10^{-5} , dropout 0.2, and gradient clipping at 1.0, with best-validation-checkpoint selection. Because pooled models converge within a few epochs and then overfit, training uses early stopping (patience 3, minimum 6 epochs) under a 12-epoch cap. Every configuration is trained under three seeds (42, 123, 2026); test metrics are reported as mean(std) over the three seeds, and Diebold-Mariano tests run on the seed-ensembled test predictions.

5. Experimental Setup

Metrics. Every configuration reports five metrics on the raw variance scale: MSE, RMSE, MAE, R^2 , and QLIKE. QLIKE, the quasi-likelihood loss standard in the realized-volatility literature [patton], penalizes under-prediction more than over-prediction and tolerates the noise in the volatility proxy:

$$\text{QLIKE} = \frac{1}{T} \sum_{t=1}^T \left(\frac{\hat{\sigma}_t^2}{\sigma_t^2} - \ln \frac{\hat{\sigma}_t^2}{\sigma_t^2} - 1 \right).$$

Training objective. All deep configurations minimize the mean squared error between the model output and the per-ticker normalized target; QLIKE and the other metrics are computed only at evaluation, after inverting the normalization to the raw variance scale, so the proportional QLIKE loss never enters the gradient.

Reporting protocol. Model selection (early-stopping, best-checkpoint) uses the validation split only; all reported results and significance tests are computed on the held-out test set; validation is used only for model selection, and no validation metrics are reported.

Significance. We complement metric comparisons with a Diebold-Mariano (DM) test on the per-observation QLIKE (and squared-error) loss series (HAC truncation lag $h-1$, Harvey-Leybourne-Newbold corrected), which tests forecast-accuracy equality directly on the held-out observations. The DM test runs on the seed-ensembled predictions (the per-observation mean forecast over seeds 42, 123, 2026).

Implementation and compute. All models are implemented in PyTorch (self-written GAT layer; no external graph library) and use a CUDA GPU. Runs used an NVIDIA GeForce RTX 4060 Laptop GPU under PyTorch 2.6 with CUDA 12.4.

6. Results

6.1 Held-out test metrics across horizons

Table 1. Held-out TEST metrics by horizon, mean(std) over three seeds (42, 123, 2026). Lower is better for MSE, RMSE, MAE, QLIKE; higher for R^2 . Bold marks the best mean per column within each horizon; the same test observations are shared across all rows of a horizon. HAR is a deterministic linear regression (std 0.00). Source: three-seed ladder_metrics.json and lstm_only_metrics.json (*.test_metrics).

$h = 1$ trading day

Config	MSE ($\times 10^{-6}$) ↓	RMSE ($\times 10^{-3}$) ↓	MAE ($\times 10^{-4}$) ↓	R^2 ↑	QLIKE ↓
HAR	4.05 (0.00)	2.014 (0.000)	5.41 (0.00)	0.8192 (0.0000)	0.4813 (0.0000)

Config	MSE ($\times 10^{-6}$) ↓	RMSE ($\times 10^{-3}$) ↓	MAE ($\times 10^{-4}$) ↓	R^2 ↑	QLIKE ↓
FULL	4.06 (0.08)	2.015 (0.020)	5.45 (0.07)	0.8189 (0.0036)	0.4853 (0.0059)
minus_graph	4.04 (0.05)	2.009 (0.012)	5.41 (0.05)	0.8200 (0.0022)	0.4797 (0.0045)
minus_gate	4.02 (0.10)	2.006 (0.025)	5.42 (0.06)	0.8206 (0.0045)	0.4830 (0.0070)
minus_news	4.01 (0.08)	2.003 (0.021)	5.42 (0.06)	0.8210 (0.0037)	0.4824 (0.0071)
LSTM-only	4.08 (0.09)	2.021 (0.022)	5.36 (0.05)	0.8179 (0.0040)	0.4791 (0.0051)

$h = 5$ trading days

Config	MSE ($\times 10^{-6}$) ↓	RMSE ($\times 10^{-3}$) ↓	MAE ($\times 10^{-4}$) ↓	R^2 ↑	QLIKE ↓
HAR	5.23 (0.00)	2.287 (0.000)	6.05 (0.00)	0.7672 (0.0000)	0.5735 (0.0000)
FULL	5.19 (0.08)	2.278 (0.018)	5.97 (0.05)	0.7691 (0.0036)	0.5768 (0.0033)
minus_graph	5.19 (0.05)	2.278 (0.011)	5.97 (0.06)	0.7691 (0.0022)	0.5781 (0.0064)
minus_gate	5.18 (0.05)	2.277 (0.012)	5.97 (0.04)	0.7693 (0.0024)	0.5757 (0.0015)
minus_news	5.21 (0.09)	2.282 (0.019)	5.97 (0.06)	0.7682 (0.0038)	0.5759 (0.0031)
LSTM-only	5.18 (0.05)	2.275 (0.012)	5.98 (0.07)	0.7696 (0.0024)	0.5772 (0.0056)

$h = 10$ trading days

Config	MSE ($\times 10^{-6}$) ↓	RMSE ($\times 10^{-3}$) ↓	MAE ($\times 10^{-4}$) ↓	R^2 ↑	QLIKE ↓
HAR	5.56 (0.00)	2.358 (0.000)	6.30 (0.00)	0.7532 (0.0000)	0.6139 (0.0000)
FULL	5.56 (0.12)	2.357 (0.026)	6.29 (0.10)	0.7534 (0.0054)	0.6441 (0.0342)
minus_graph	5.50 (0.08)	2.345 (0.017)	6.35 (0.03)	0.7560 (0.0035)	0.6179 (0.0033)
minus_gate	5.49 (0.03)	2.344 (0.006)	6.34 (0.10)	0.7562 (0.0012)	0.6225 (0.0033)
minus_news	5.55 (0.12)	2.356 (0.025)	6.31 (0.14)	0.7535 (0.0053)	0.6595 (0.0533)
LSTM-only	5.49 (0.05)	2.342 (0.011)	6.33 (0.04)	0.7566 (0.0023)	0.6189 (0.0043)

$h = 22$ trading days

Config	MSE ($\times 10^{-6}$) ↓	RMSE ($\times 10^{-3}$) ↓	MAE ($\times 10^{-4}$) ↓	R^2 ↑	QLIKE ↓
HAR	6.02 (0.00)	2.453 (0.000)	6.56 (0.00)	0.7303 (0.0000)	0.6742 (0.0000)
FULL	6.11 (0.03)	2.472 (0.006)	6.54 (0.07)	0.7262 (0.0014)	0.7053 (0.0066)
minus_graph	6.15 (0.14)	2.479 (0.028)	6.53 (0.02)	0.7245 (0.0062)	0.6956 (0.0140)
minus_gate	6.22 (0.18)	2.493 (0.037)	6.62 (0.05)	0.7214 (0.0082)	0.7058 (0.0079)
minus_news	6.19 (0.09)	2.489 (0.018)	6.60 (0.01)	0.7225 (0.0041)	0.7097 (0.0107)
LSTM-only	6.09 (0.02)	2.468 (0.004)	6.54 (0.05)	0.7271 (0.0009)	0.7003 (0.0067)

Reading across horizons: on the squared-error metrics (MSE, RMSE, R^2) the configuration means fall within about 1% of one another and the best row alternates (minus_news at h1, minus_gate/LSTM-only at h5/h10, HAR at h22); on MAE the differences are similarly small and the lowest mean varies by horizon (LSTM-only at h1, four configurations tied near 5.97 at h5, FULL at h10, minus_graph at h22); on QLIKE, HAR has the lowest mean at h5, h10, and h22 while the price-only LSTM-only has the lowest mean at h1. The reported standard deviations are small relative to the between-configuration gaps except at h10, where the FULL and minus_news QLIKE means carry larger seed dispersion (std 0.034 and 0.053). The remaining subsections isolate each component (Section 6.2), compare FULL against HAR (Section 6.3), test significance (Section 6.4), and evaluate forecast calibration (Section 6.5).

6.2 Component contributions across horizons

Table 2. Component contribution on held-out test QLIKE, $\text{effect}(X) = \text{QLIKE}(\text{FULL}) - \text{QLIKE}(\text{FULL} - X)$, **three-seed mean.** Negative = removing X raised QLIKE, i.e. X helped. Source: three-seed ladder_metrics.json (leave_one_out_effects).

Horizon	effect(graph)	effect(gate)	effect(news)
1	+0.00566	+0.00235	+0.00297
5	−0.00130	+0.00105	+0.00088
10	+0.02617	+0.02159	−0.01541
22	+0.00979	−0.00041	−0.00432

Reading: effect(graph) is positive at h1, h10, and h22 (removing the GAT branch lowers QLIKE there) and marginally negative at h5 (−0.0013). effect(gate) is positive at h1, h5, and h10 and marginally negative at h22 (−0.0004). effect(news) is positive at h1 and h5 and negative at h10 (−0.0154) and h22 (−0.0043), so removing the news branch raises QLIKE only at the longer horizons. These are QLIKE effects; the corresponding MSE, RMSE, and R^2 differences among the same configurations are

within about 1% (Table 1). The magnitudes at h10 are the largest, but the seed dispersion at h10 (Table 1) is also the largest; the Diebold-Mariano test (Section 6.4) determines which differences are significant on the seed-ensembled predictions.

6.3 FULL versus HAR across horizons

Table 3. FULL vs HAR, held-out test, all four horizons, mean(std) over three seeds. Columns pair each metric HAR-then-FULL; bold marks the better mean of the pair. Source: three-seed ladder_metrics.json (rungs.FULL.test_metrics, rungs.HAR.test_metrics).

Horizon	HAR QLIKE	FULL QLIKE	HAR R^2	FULL R^2	HAR RMSE ($\times 10^{-3}$)	FULL RMSE ($\times 10^{-3}$)
1	0.4813 (0.0000)	0.4853 (0.0059)	0.8192 (0.0000)	0.8189 (0.0036)	2.014 (0.000)	2.015 (0.020)
5	0.5735 (0.0000)	0.5768 (0.0033)	0.7672 (0.0000)	0.7691 (0.0036)	2.287 (0.000)	2.278 (0.018)
10	0.6139 (0.0000)	0.6441 (0.0342)	0.7532 (0.0000)	0.7534 (0.0054)	2.358 (0.000)	2.357 (0.026)
22	0.6742 (0.0000)	0.7053 (0.0066)	0.7303 (0.0000)	0.7262 (0.0014)	2.453 (0.000)	2.472 (0.006)

Reading: on RMSE and R^2 the full model and HAR are within their seed dispersion at h1, h5, and h10, and HAR is marginally ahead at h22 (MSE and MAE follow the same pattern in Table 1). On QLIKE, HAR has the lower mean at all four horizons, with the gap widening from h1 (0.0040) to h22 (0.0311). The Diebold-Mariano table (Table 4) tests these gaps across loss families on the seed-ensembled predictions: on QLIKE the full model shows no significant difference from HAR at h1, h5, and h10 and HAR is significantly lower at h22; on squared error there is no significant difference at any horizon; on absolute error the full model is significantly lower at h5.

6.4 Diebold-Mariano significance across metrics

Table 4. Diebold-Mariano (HLN) on held-out test, seed-ensembled (42, 123, 2026), per horizon and per loss family. Each cell is the HLN-corrected DM statistic with its two-sided p -value; **bold** marks $p < 0.05$. A **negative** statistic means FULL has the lower loss; a **positive** statistic means the comparator has the lower loss. DM is run on three per-observation loss families: QLIKE, squared error (SE; the MSE/RMSE/ R^2 family), and absolute error (AE; MAE). HAC truncation lag $h-1$; n per horizon is 14,596 (h1), 14,464 (h5), 14,299 (h10), 13,903 (h22).

$h = 1$ trading day

Comparison	QLIKE	SE (MSE/RMSE/ R^2)	AE (MAE)
FULL vs HAR	+0.73 ($p = 0.46$)	-0.69 ($p = 0.49$)	+0.88 ($p = 0.38$)
FULL vs minus_graph	+8.70 ($p < 0.001$)	-0.21 ($p = 0.83$)	+2.66 ($p = 0.01$)

Comparison	QLIKE	SE (MSE/RMSE/ R^2)	AE (MAE)
FULL vs minus_gate	+6.78 ($p < 0.001$)	+2.37 ($p = 0.02$)	+6.24 ($p < 0.001$)
FULL vs minus_news	+8.04 ($p < 0.001$)	+3.93 ($p < 0.001$)	+5.35 ($p < 0.001$)
FULL vs LSTM- only	+6.07 ($p < 0.001$)	-1.78 ($p = 0.07$)	+6.77 ($p < 0.001$)

$h = 5$ trading days

Comparison	QLIKE	SE (MSE/RMSE/ R^2)	AE (MAE)
FULL vs HAR	-0.34 ($p = 0.74$)	-1.03 ($p = 0.30$)	-4.20 ($p < 0.001$)
FULL vs minus_graph	+1.29 ($p = 0.20$)	+0.27 ($p = 0.79$)	-0.58 ($p = 0.56$)
FULL vs minus_gate	+0.93 ($p = 0.35$)	+0.23 ($p = 0.82$)	-0.53 ($p = 0.60$)
FULL vs minus_news	+4.45 ($p < 0.001$)	-0.94 ($p = 0.35$)	-0.94 ($p = 0.35$)
FULL vs LSTM- only	+2.58 ($p = 0.01$)	+0.65 ($p = 0.52$)	-0.49 ($p = 0.62$)

$h = 10$ trading days

Comparison	QLIKE	SE (MSE/RMSE/ R^2)	AE (MAE)
FULL vs HAR	+1.28 ($p = 0.20$)	-0.20 ($p = 0.84$)	-1.07 ($p = 0.28$)
FULL vs minus_graph	+0.80 ($p = 0.42$)	+2.25 ($p = 0.02$)	-5.20 ($p < 0.001$)
FULL vs minus_gate	+0.08 ($p = 0.94$)	+2.06 ($p = 0.04$)	-4.34 ($p < 0.001$)
FULL vs minus_news	+1.10 ($p = 0.27$)	+1.30 ($p = 0.19$)	-2.08 ($p = 0.04$)
FULL vs LSTM- only	+0.56 ($p = 0.58$)	+2.62 ($p = 0.01$)	-4.74 ($p < 0.001$)

$h = 22$ trading days

Comparison	QLIKE	SE (MSE/RMSE/ R^2)	AE (MAE)
FULL vs HAR	+5.10 ($p < 0.001$)	+1.09 ($p = 0.27$)	-1.21 ($p = 0.22$)
FULL vs minus_graph	+5.88 ($p < 0.001$)	+0.08 ($p = 0.94$)	+2.50 ($p = 0.01$)
FULL vs minus_gate	+1.56 ($p = 0.12$)	-1.52 ($p = 0.13$)	-3.22 ($p < 0.001$)

Comparison	QLIKE	SE (MSE/RMSE/ R^2)	AE (MAE)
FULL vs minus_news	-3.14 ($p < 0.001$)	-1.90 ($p = 0.06$)	-4.25 ($p < 0.001$)
FULL vs LSTM- only	+2.23 ($p = 0.03$)	+0.75 ($p = 0.45$)	+0.25 ($p = 0.80$)

Reading (all loss families): For FULL vs HAR, the QLIKE loss differs significantly only at h22 (HAR lower, $p < 0.001$) with no significant difference at h1, h5, or h10; the squared-error loss shows no significant difference at any horizon; the absolute-error loss differs significantly only at h5 (FULL lower, $p < 0.001$). No single model has the lower loss across all three families. For the component removals, significance depends on the loss and the horizon: removing the GAT branch is significant on QLIKE at h1 and h22 and on the squared-error loss at h10 only; removing the per-ticker gate is significant on QLIKE at h1; removing the news branch is significant on QLIKE at h1, h5, and h22. The price-only LSTM backbone has the lower or equal loss relative to the full model on QLIKE at h1, h5, and h22. Across the three loss families, no configuration is uniformly favored over HAR.

6.5 Mincer-Zarnowitz forecast efficiency

The Diebold-Mariano tests compare forecast *accuracy*; the Mincer-Zarnowitz (MZ) regression evaluates forecast *calibration/bias*, a complementary property. For each rung and horizon we regress the realized variance y on the seed-ensembled forecast x ,

$$y = a + bx + e,$$

and test the joint efficiency hypothesis $H_0 : a = 0$ and $b = 1$ with a Wald statistic $W = (\hat{\theta} - \theta_0)'V^{-1}(\hat{\theta} - \theta_0)$ ($\hat{\theta} = (a, b)$, $\theta_0 = (0, 1)$, $V = \hat{\sigma}^2(X'X)^{-1}$), which is χ^2_2 under H_0 . A well-calibrated forecast has an intercept near zero, a slope near one, and a large joint p -value; a slope above one indicates the forecast under-responds to realized variance (an attenuation bias), and a slope below one indicates the opposite.

Table 5. Mincer-Zarnowitz regression of realized on forecast variance, seed-ensembled (42, 123, 2026), per horizon. Intercept a in $\times 10^{-6}$ (target scale $\approx 10^{-4}$), slope b , regression R^2 , and the joint Wald statistic with its χ^2_2 p -value ($H_0 : a = 0, b = 1$). Bold marks, per horizon and per column, the value closest to calibration (smallest $|a|$, slope closest to 1, highest R^2 , smallest Wald). n per horizon is 14,596 (h1), 14,464 (h5), 14,299 (h10), 13,903 (h22). Source: `mz_report.py` over the seed-ensembled per-rung predictions_test.json dumps.

$h = 1$ trading day

Rung	$a (\times 10^{-6})$	b	R^2	Wald	joint p
HAR	-13.95	1.0611	0.8220	234.6	<0.001
FULL	-14.69	1.0461	0.8226	136.9	<0.001
minus_graph	-6.50	1.0410	0.8220	110.5	<0.001
minus_gate	-12.91	1.0522	0.8241	176.3	<0.001
minus_news	-12.53	1.0445	0.8244	129.7	<0.001

Rung	$a (\times 10^{-6})$	b	R^2	Wald	joint p
LSTM-only	+5.42	1.0521	0.8210	179.2	<0.001

$h = 5$ trading days

Rung	$a (\times 10^{-6})$	b	R^2	Wald	joint p
HAR	−10.58	1.0793	0.7717	280.5	<0.001
FULL	+6.82	1.0793	0.7744	293.1	<0.001
minus_graph	+12.29	1.0618	0.7730	185.0	<0.001
minus_gate	+10.18	1.0679	0.7734	219.9	<0.001
minus_news	+11.46	1.0693	0.7726	228.2	<0.001
LSTM-only	+12.75	1.0579	0.7732	164.3	<0.001

$h = 10$ trading days

Rung	$a (\times 10^{-6})$	b	R^2	Wald	joint p
HAR	−13.81	1.1009	0.7600	405.0	<0.001
FULL	+10.77	1.0809	0.7587	277.4	<0.001
minus_graph	−14.51	1.0791	0.7616	260.1	<0.001
minus_gate	−7.98	1.0754	0.7614	239.8	<0.001
minus_news	+7.46	1.0779	0.7590	258.0	<0.001
LSTM-only	−11.18	1.0791	0.7619	261.8	<0.001

$h = 22$ trading days

Rung	$a (\times 10^{-6})$	b	R^2	Wald	joint p
HAR	−29.24	1.1388	0.7420	627.4	<0.001
FULL	−15.96	1.1610	0.7418	822.0	<0.001
minus_graph	−3.41	1.1462	0.7396	694.7	<0.001
minus_gate	−15.93	1.1474	0.7361	684.4	<0.001
minus_news	−26.38	1.1757	0.7409	944.4	<0.001
LSTM-only	−1.63	1.1260	0.7386	531.9	<0.001

Reading: the intercepts are near zero at every horizon (all $|a| \leq 30 \times 10^{-6}$ against a target on the 10^{-4} scale), so the level bias is small and the calibration gap is carried by the slope. Every slope exceeds one and the departure from one increases with the horizon, from about 1.04–1.06 at h1 to about 1.13–1.18 at h22, indicating that the forecasts increasingly under-respond to realized variance at longer horizons. The joint hypothesis $a = 0, b = 1$ is rejected for every rung at every horizon ($p < 0.001$), consistent with the large sample size ($n \approx 13,900$ – $14,600$). By the Wald magnitude, the configuration closest to MZ efficiency is minus_graph at h1, LSTM-only at h5 and h22, and minus_gate at h10; no single configuration is closest to calibration across all four horizons. MZ measures calibration/bias and is complementary to the QLIKE/MSE accuracy comparisons in Sections 6.1–6.4.

7. Discussion

What each component contributes. Table 2 reports each component’s marginal QLIKE contribution, and Table 4 its significance on the seed-ensembled predictions; on the MSE, RMSE, and R^2 metrics the same configurations differ by about 1% or less (Table 1). For the GAT branch, removing it significantly lowers QLIKE at h1 and h22 ($p < 0.001$) with no significant difference at h5 or h10, so the directed lead-lag edge — the design chosen specifically to avoid the market-factor redundancy of a correlation edge — yields no significant out-of-sample QLIKE improvement at any horizon. This is consistent with the earlier correlation-edge finding: redirecting the edge to a predictive volume→volatility relation and giving the attention raw features does not produce a QLIKE improvement. For the news branch, removing it significantly lowers QLIKE at h1 and h5 and significantly raises it at h22 ($p = 0.0017$), so the news branch lowers QLIKE only at h22. For the gate, removing it significantly lowers QLIKE at h1 and shows no significant difference at the other horizons. Relative to HAR, the full model shows no significant QLIKE difference at h1, h5, and h10 and has significantly higher QLIKE at h22 (Table 4).

The price-only backbone relative to the full stack. The LSTM-only reference (a shared pooled price LSTM with no news, no gate, no graph) has three-seed test QLIKE 0.4791 / 0.5772 / 0.6189 / 0.7003 at h1 / h5 / h10 / h22, and the seed-ensembled DM test finds it has significantly lower or equal QLIKE relative to the full model at h1, h5, and h22 (no significant difference at h10). Relative to HAR, both LSTM-only and the full model show no significant QLIKE advantage at h1–h10 and higher QLIKE than HAR at h22, so the deep temporal model matches HAR at the short horizons and does not overtake the linear baseline at the long horizon.

Forecast calibration. The Mincer-Zarnowitz regression (Section 6.5) finds near-zero intercepts and slopes above one at all four horizons, with the joint efficiency hypothesis rejected for every configuration ($p < 0.001$) and the slope departure from one growing with the horizon, and no single configuration is closest to calibration across all four horizons.

Overall reading. Across three seeds and all five metrics, no configuration consistently or significantly outperforms HAR. On MSE, RMSE, MAE, and R^2 the configurations differ by about 1% or less, within seed dispersion, at every horizon; on QLIKE, HAR shows no significant difference from the full model at h1, h5, and h10 and a significantly lower value at h22, and the price-only LSTM backbone matches or has a significantly lower QLIKE than the full model at h1, h5, and h22. The graph, gate, and news components do not provide a consistent significant improvement across horizons on any metric.

Relation to the literature. A null or negligible graph effect aligns with the best-controlled published GNN-vs-HAR study, GNNHAR on DJIA-30, where multi-hop graph spillover gave no clear advantage under a Model Confidence Set and the gains came from nonlinearity and a QLIKE training loss rather than the graph [10], and with the broader finding that a well-specified HAR is hard to beat on a limited information set [14].

Levers not exercised here. The literature identifies a QLIKE training objective and intraday-derived realized variance as the decisive levers for beating HAR [10]; this study trains on MSE and forecasts a daily range-based variance, so the ablation isolates architecture rather than objective or estimator. These remain future work.

8. Limitations

First, the 33-ticker universe is a fixed, point-in-time VN30-like set that excludes two short-history current members (BSR, VPL), so it is not the live index. Second, the study covers a single market at daily frequency; the results may not transfer to higher frequencies or to markets with balanced listing histories, and every rigorous published GNN-beats-HAR result relies on intraday-derived realized variance this daily-OHLCV panel does not have. Third, the reported numbers aggregate three seeds (42, 123, 2026) with seed-ensembled Diebold-Mariano tests; a five-seed extension and a Model Confidence Set are possible future work to further tighten the significance estimates. Fourth, training uses an MSE objective rather than the QLIKE objective the literature credits for closing the HAR gap.

9. Conclusion

We propose a parallel multi-branch model for multi-horizon VN30 Parkinson-variance forecasting that fuses a price LSTM, a real multi-head GAT over a directed volume→volatility lead-lag edge, and a gated PhoBERT news branch, with the GAT consuming raw node features to match the edge semantics. We evaluate it with an ablation that retrains one variant per removed component and attributes each component’s contribution as the change it causes in held-out QLIKE, against a strong HAR baseline and with seed-ensembled Diebold-Mariano significance, at horizons 1, 5, 10, and 22 trading days over three seeds. Across all five metrics no configuration consistently or significantly outperforms HAR: on MSE, RMSE, MAE, and R^2 the configurations differ by about 1% or less at every horizon, and on QLIKE the full model shows no significant difference from HAR at h1, h5, and h10 and a significantly higher value at h22. The ablation removals give mixed, mostly short-horizon QLIKE effects — removing the GAT branch significantly lowers QLIKE at h1 and h22, removing the per-ticker gate at h1, and removing the news branch at h1 and h5 while raising it at h22 — and a price-only LSTM backbone has significantly lower or equal QLIKE relative to the full model at h1, h5, and h22. A Mincer-Zarnowitz regression finds near-zero intercepts and slopes above one at every horizon, with the joint efficiency hypothesis rejected for every configuration. For a Vietnamese emerging market, the study shows how to build and honestly evaluate a directed-spillover graph-attention news forecaster against the field-standard HAR.

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